AuthService Flowchart
AccountController Sequence Diagram

%%
AuthService Flowchart
Register(fullName, email, password, phone, role, specialty?)
Start
EmailExists(email)?
No
Create User (hash password); Add to Users; SaveChanges
Role?
Admin/Other
No subtype record (Admin/other)
Worker
Add Worker with Specialty; IsVerified = false; Availability = Available
Add Wallet for Worker with Balance = 0
SaveChanges
Return (true, "Registration successful.")
Customer
Add Customer with TotalRequests = 0
Yes
Return (false, "Email is already registered.")
Login(email, password)
Start
GetByEmail(email)
User found?
Yes
No
PasswordHash empty/null?
No
BCrypt.Verify(password, PasswordHash)
Valid?
No
Return null
Yes
Yes
Return user
FindByGoogleId(googleId)
Start
GetByGoogleId(googleId)
Return User or null
FindByEmail(email)
Start
GetByEmail(email)
Return User or null
LinkGoogleId(user, googleId)
Start
Set user.GoogleId = googleId
Update Users
SaveChanges
End
RegisterGoogleUser(fullName, email, googleId, phone, role, specialty?)
Start
EmailExists(email)?
No
Create User (PasswordHash=""); GoogleId set; Add to Users; SaveChanges
Role?
Admin/Other
No subtype record (Admin/other)
Worker
Add Worker with Specialty; IsVerified = false; Availability = Available
Add Wallet for Worker with Balance = 0
SaveChanges
Return (true, "Google registration successful.")
Customer
Add Customer with TotalRequests = 0
Yes
Return (false, "Email already registered.")
AccountController Sequence Diagram
POST Login(model)
Start
User -> AccountController: Submit credentials
ModelState valid?
Yes
AuthService.Login
UserRepository.GetByEmail
Password checks + BCrypt.Verify
Credentials valid?
No
Return Login view with "Invalid email or password."
Yes
CookieAuth.SignInAsync (SignInUser)
RedirectByRole
Redirect to Dashboard/Profile/Home based on role
End
No
Return Login view with validation errors
POST Register(model)
Start
User -> AccountController: Submit registration
ModelState valid?
Yes
Role=Worker AND Specialty missing?
Yes
Return view: Specialty required
No
AuthService.Register -> EmailExists? -> Create & Save User
Create subtype records by role (Customer/Worker+Wallet)
Success?
Yes
AuthService.FindByEmail
CookieAuth.SignInAsync (SignInUser)
RedirectByRole
Redirect by role
End
No
Return Register view with error message
No
Return Register view with validation errors
GET GoogleLogin()
Start
User -> AccountController: Request Google login
Create AuthenticationProperties (RedirectUri=GoogleCallback)
Challenge Google authentication scheme
End
GET GoogleCallback()
Start
CookieAuth.AuthenticateAsync
Authenticated?
Yes
Extract googleId, email, name claims
AuthService.FindByGoogleId
User found by GoogleId?
Yes
No
AuthService.FindByEmail
User exists after linking?
Yes
AuthService.LinkGoogleId
CookieAuth.SignInAsync (SignInUser)
RedirectByRole
Redirect by role
End
No
Return CompleteProfile view (prefilled)
No
Redirect to Login
POST CompleteProfile(model)
Start
User -> AccountController: Submit completed profile
ModelState valid?
Yes
Role=Worker AND Specialty missing?
Yes
Return view: Specialty required
No
AuthService.RegisterGoogleUser -> EmailExists? -> Create & Save User
Success?
Yes
AuthService.FindByEmail
CookieAuth.SignInAsync (SignInUser)
RedirectByRole
Redirect by role
End
No
Return view with error message
No
Return view with validation errors
POST Logout()
Start
User -> AccountController: Logout request
CookieAuth.SignOutAsync
Redirect to Login
End